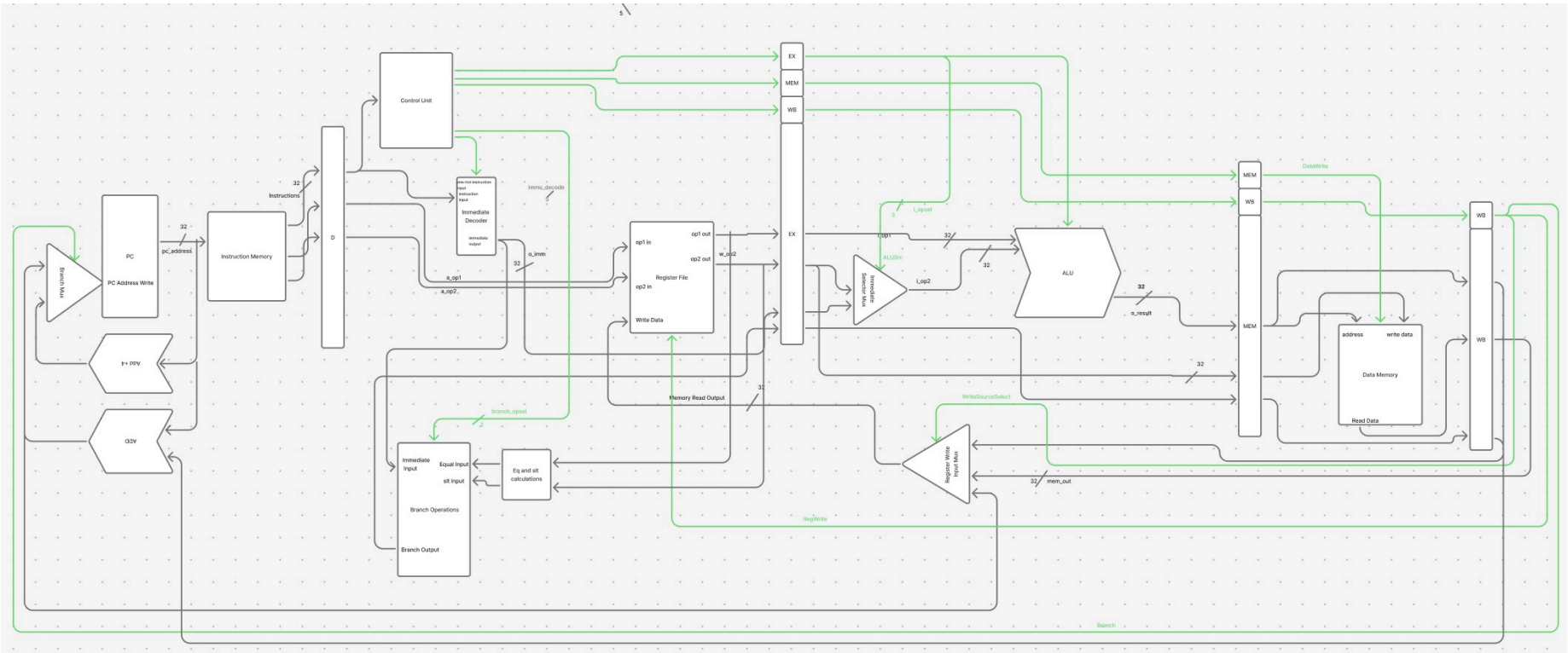
5
Control Unit
EX
MEM
WB
Instructions
32
D
Immediate Decoder
immediate output
immu_decode
7
PC
PC Address Write
pc_address
Instruction Memory
Branch Mux
ADD +4
ADD
a_imm
a_op1
a_op2
Register File
op1 in
op2 in
Write Data
op1 out
op2 out
w_op2
Memory Read Output
branch_opsel
2
Immediate Input
Equal Input
slt Input
Branch Operations
Branch Output
Eq and slt calculations
ALUOperand Selector Mux
ALUSrc
i_op1
i_op2
i_opsel
3
ALU
o_result
Register Write Input Mux
WriteSourceSelect
mem_out
RegWrite
Branch
DataWrite
address
write data
Data Memory
Read Data